Smart Clinic AI Assistant

Project Info Form

DEPI — Digital Egypt Pioneers Initiative | Track: AI & Data Science

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Project name: Smart Clinic AI Assistant

Project Idea: An n8n-based automation system for a single-doctor clinic that handles patient communication end-to-end via a Telegram chatbot. Patients choose from button menus to ask about the clinic or book an appointment, with real-time Google Calendar availability checks to prevent double-booking. An AI agent answers free-text questions, restricted to clinic information; it never books appointments and gives no medical advice. The system also sends 24-hour interactive appointment reminders, generates an automated daily financial report (calculated in code, not by the LLM), and collects post-visit feedback that an LLM classifies by sentiment. Anything that must be exact (availability, bookings, money) is decided by code. This reduces manual staff work and gives the doctor accurate, up-to-date visibility into clinic performance.

Github repo URL: https://github.com/nhahub/NHA-5-201

Tools, Libraries and APIs

- n8n (core automation/orchestration engine)
- Telegram Bot API
- Google Calendar API
- Google Sheets API
- Gmail API
- LLM API (AI agent for free-text answers, and feedback sentiment classification)

Project Team Members

- Ahmed Hossam Abdullah
- Mohamed Alaa Ibrahim
- Ziad Ahmed Fathy
- Ahmed Mossa Anwar

Team Leader

- Ahmed Khaled Abdelfattah

Final Project Deliverable: A working n8n automation system (not a standalone app) — a set of deployed, interconnected n8n workflows exposed to patients through a Telegram bot, demoed live rather than packaged as an api/mobile/web app.

Project Pipeline:

1 Patient sends a message via Telegram.

2. Button menu routes the patient to Inquiry or Booking. Free-text messages that are not part of a booking step are answered by the AI agent (clinic information only).
3. Booking path checks Google Calendar for availability (code-based) — single doctor, 20-minute slots. The chosen slot is remembered per chat, so name and age can be sent as a normal message, and the slot is checked again at confirmation to prevent double-booking.
4. Confirmed data (patient, appointment, chat_id) is logged to Google Sheets — the shared data store.
5. From Sheets, three scheduled jobs run independently: Reminders (sent 24h before each appointment); Financial report (daily revenue and case count, calculated in code and emailed to the doctor); Feedback (LLM sentiment analysis on post-visit patient responses).

(Recreate this flow as a diagram in diagrams.net per the submission requirement — the stages above are the exact content to lay out.)

Dataset source: No fixed external dataset — the system operates on live data generated during operation (bookings, appointments, and feedback captured through Telegram/Calendar/Sheets), not pre-collected training data.

Timeline Deliverables

Week 5 & 6:

Weeks deliverables (project-wise):

- **Implementation complete (due 30 Nov 2026):**
- Google Sheets logging for every confirmed booking (appointment_id, chat_id, name, age, date, time, status).
- Reminder workflow: scheduled run 24 hours before each appointment. Confirm updates the status; Cancel deletes the Calendar event and updates the status; Reschedule marks "Needs Rescheduling" and returns the patient to booking.
- Daily financial report: total income and case count calculated by a Code node from the Daily Ledger and emailed to the doctor via Gmail.
- Feedback workflow: request sent after attended visits, LLM sentiment classification (Positive, Neutral, Negative), results stored in the Feedback sheet.
- AI agent finalized: restricted system prompt (clinic information only, no booking, no medical advice) and a fallback to the menu if the AI service fails.
- Security: bot token replaced, all credentials kept only in n8n, and the workflow JSON exported to the repository (workflows folder) without secrets.
- **Testing and QA:** run the test plan from the System Analysis & Design document (T1 to T16), log bugs as GitHub Issues, fix and retest.

- **Documentation:** README updated (installation, configuration, run guide), User Manual, Technical Documentation draft; Proposal, README, and Info Form updated to include the AI agent.
- **Final presentation preparation (due 4 Dec 2026):** presentation slides, project infographic, and an optional demo video.

Team Members' Roles:

- Ahmed Khaled Abdelfattah (Team Leader) role: integration review and final sign-off, README and repository organization (workflows folder), presentation and infographic, deadline follow-up
- Ahmed Hossam Abdullah role: Google Sheets logging and the reminder workflow
- Mohamed Alaa Ibrahim role: AI agent finalization, daily financial report workflow, feedback workflow, and token security
- Ziad Ahmed Fathy role: executing the test plan (T1 to T16) and writing bug reports
- Ahmed Mossa Anwar role: User Manual, technical documentation, and presentation slides